

Raghav Senthil Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, BuildIllinois, Innovation LLC, Agentic AI

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Collaborated in an *Agile* team of **10 engineers** to design, develop and test the *Python (FastAPI)* search engine behind a SaaS research tool for high school debaters, deployed on the *AWS* cloud with **10K+ site visitors**
- Indexed **600K+ documents** on *Elasticsearch*, a big data platform tool, keeping search latency below **1 second**
- Solved a complex retrieval problem with a *Python* algorithm that validates source URLs and filters **IQR** outliers
- Improved solution performance by de-duplicating the search index, where **30%** of documents were identical copies

Product Development Intern | Jetson

January 2025 – February 2025

- Improved user experience by prototyping a *Figma* onboarding redesign that cut the flow from **12 pages to 5**
- Evaluated 3 experimental app features across **7 rounds of UI/UX testing** in an app used by **100K+ users**
- Articulated findings and proposed solutions to the CEO in a **5-page report** on improving user engagement

Data Analytics Intern | STEM E

June 2024 – July 2024

- Performed data analysis on **200+ TikTok posts** from 3 years using *Python (pandas)*, *Power BI* and *Excel*
- Transformed raw engagement data into a key finding: “science fun facts” drove 69.3% of views from 7% of content
- Proposed a solution to the Executive Director’s engagement challenge that lifted engagement **24% (8K+ views)**

PROJECTS

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Developed a generative AI solution that pairs a **Gemma Voice LLM** with **4 Random Forest models** to turn a school’s live energy, water and waste data into spoken, actionable steps on a *Next.js* dashboard for its users
- Deployed the models behind a *Python FastAPI* back end that ingests live data streams from 3 edge AI tools
- Led 5 peers to research and propose a school sustainability solution. **USAII Global Hackathon Finalist**

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Evaluated generative AI capabilities for **Kamion (YC S22)**, iteratively testing a **Claude Sonnet 4.6** vision-language agent against real truck photos until it read **10 images in a row** correctly on all 3 features
- Designed and configured a *JSON schema* so the agent returns each truck’s condition, brand, model and year
- Transformed **15,000** scraped listing images into a clean dataset by removing duplicates and non-USD prices
- Selected a *quantile regression* model to price each truck from its features, hitting **87% accuracy** over 30 samples

EcoBin | [Research Paper](#) | [Source Code](#)

March 2026 – May 2026

- Selected and trained a two-stage neural network (*EfficientNetV2-S*, **4.39M** parameters) in *Python* with *PyTorch* on **15K+ images**, then validated it at **96.13% accuracy** for routing waste into 4 distinct disposal categories
- Built a data transformation pipeline (*U2-Net* segmentation, alpha-compositing) generating **9K** synthetic images
- Deployed the classifier in a smart bin prototype on an *Arduino Uno* that sorts each item in **under 2 seconds**
- Documented the research in an *arXiv* paper and 2 *Kaggle* datasets. **2nd at Arizona State Science Fair**

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Applied AI engineering to deploy a **Claude Sonnet 4.5 RAG** agent grounded in per-session *Supermemory* vectors
- Built a data pipeline that transforms papers from 2 APIs (*OpenAlex*, *Semantic Scholar*) into *PostgreSQL* records
- Prototyped an interactive *React Flow* graph showing where a paper’s citations land across **26 research fields**

SKILLS

Programming Languages: Python, SQL, JavaScript, Java, C++, HTML/CSS

Concepts/Practices: Generative AI, LLMs, RAG, AI Agents, Prompt Engineering, Data Pipelines, ETL, Agile

AI/ML: Claude API, Gemma 4, Hugging Face, PyTorch, scikit-learn, TensorFlow, Keras, OpenCV, Vercel AI SDK

Frameworks/Platforms: AWS, Google Cloud, FastAPI, Elasticsearch, Next.js, React, PostgreSQL, Supabase, Flask

Data & Analytics: Microsoft Excel, pandas, NumPy, Power BI, A/B Testing, Hootsuite